

Python OOP & Exception Handling Practice Questions

1. Create a class Student with attributes name and marks. Create objects and display details.
2. Create a class Laptop that initializes brand and price using a constructor.
3. Create a class Rectangle with length & width and a method to calculate area.
4. Create a class Employee with a class variable company_name and instance variable employee_name.
5. Create a class Calculator with a static method to add two numbers.
6. Create class Animal with method sound(). Create class Cat that inherits and overrides the method.
7. Create multilevel inheritance with Person → Employee → Manager, each adding properties.
8. Create multiple inheritance: Father (showFather), Mother (showMother), Child inherits both.
9. Create class BankAccount with withdraw(). Override it in SavingsAccount to maintain minimum balance.
10. Create class Account with private variable __balance and methods deposit() & withdraw().
11. Create classes Circle and Square with method area(). Use a loop to call both (polymorphism).
12. Create abstract class Vehicle with abstract method start(). Implement in Bike and Car.
13. Take two numbers and handle division by zero.
14. Take integer input and handle invalid input errors.
15. Handle ValueError and ZeroDivisionError in a calculator program.
16. Open a file and ensure it closes properly using finally.
17. Create custom exception InsufficientBalanceError for withdrawal exceeding balance.
18. Check voting eligibility. Raise exception if age < 18.
19. Access list element by user index and handle index errors.
20. Write a program using try-except-else for division and print result if no error.
21. What is difference between Machine Learning and Deep Learning, include algorithms.